No universal weighted Sobolev–SVD tail bound

A smooth rank-two Fourier counterexample

Autonomous mathematical discovery run `fa_banach_001`

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Status. Candidate full negative answer, likely valid pending expert review.

1 Source question

Ali and Nouy compare the ordinary L^2 singular value decomposition

$$u = \sum_{k\geq 1} \sigma_k^{00} \psi_k \otimes \phi_k$$

with the SVD obtained by viewing the same kernel as a Hilbert–Schmidt operator $L^2(\Omega_2) \to H^1(\Omega_1)$; its singular values are denoted σ_k^{10}. Immediately after Proposition 2.1 of [1], on PDF page 3, they ask whether one can bound

$$\sum_{k>r} (\sigma_k^{00})^2 \|\psi_k\|^2_{H^1(\Omega_1)} \lesssim \sum_{k>r} (\sigma_k^{10})^2 \gamma(k) \tag{1}$$

for some sequence $\gamma(k)$.

> Analogously for the second statement. □
>
> Note that the upper bounds in Proposition 2.1 do not necessarily hold component-wise, i.e., the inequalities $\sigma_k^{10} \leq \sigma_k^{00}\|\psi_k\|_1$, do not hold in general. This is due to the fact that when estimating the injective norm
>
> $$\left\| \sum_{k=r+1}^{\infty} \sigma_k^{00} \psi_k \otimes \phi_k \right\|_{\vee(H^1(\Omega_1), L^2(\Omega_2))},$$
>
> the functions ψ_k are not orthonormal in $H^1(\Omega_1)$ and the sequence $\{\sigma_k^{00}\|\psi_k\|_1\}_{k\in\mathbb{N}}$ is not necessarily decreasing.
>
> Naturally, we can ask whether we can derive a bound of the sort
>
> $$\sum_{k=r+1}^{\infty} (\sigma_k^{00})^2 \|\psi_k\|_1^2 \lesssim \sum_{k=r+1}^{\infty} (\sigma_k^{10})^2 \gamma(k),$$
>
> for some sequence $\gamma(k)$. Though we do not believe this is possible without further assumptions, we can nonetheless improve the bounds. This indicates that indeed the quantities σ_k^{10} and $\sigma_k^{00}\|\psi_k\|_1$ are closely related. This will later be confirmed by numerical observations.
>
> **Theorem 2.2.** *Let $u \in H^1(\Omega)$ and assume the L^2-SVD $u = \sum_{k=1}^{\infty} \sigma_k^{00} \psi_k \otimes \phi_k$ converges in $H^1(\Omega)$. Then, we have*
>
> $$\sigma_r^{10} = \|\psi_r^1\|_0^{-1/2} \left(\sum_{k=1}^{\infty} (\sigma_k^{00})^4 |\langle \psi_r^1, \psi_k^0 \rangle_1|^2 \right)^{1/4} \geq \sigma_r^{00} \left(\frac{\langle \psi_r^1, \psi_r \rangle_1}{\|\psi_r^1\|_0} \right)^{1/2},$$

We use the only nontrivial uniform interpretation: the finite-valued sequence γ and the implicit constant are fixed independently of u. If either may depend arbitrarily on u, the assertion carries no uniform content.

2 Proof intuition

Mix one very high Fourier mode with the constant mode by a 45-degree rotation. The resulting two functions remain orthogonal in L^2, so they can be prescribed as the unique left singular functions of an L^2-SVD. In H^1, however, almost all of their growing derivative energy points in the same direction.

The strong-norm SVD detects that common direction and places the growing energy in its first singular value. Its second singular value stays bounded. After truncating the first term, the source's weighted L^2 tail still contains half of the high-frequency derivative energy, while the proposed right side contains only the bounded second strong-norm singular value. The failure already occurs at index two, so no growth of later weights can help.

3 Counterexample

Equip $H^1((0, 2\pi))$ with

$$\langle f, g\rangle_{H^1} = \int_0^{2\pi} (fg + f'g')\, dx.$$

Let

$$e_0(x) = (2\pi)^{-1/2}, \qquad e_N(x) = \pi^{-1/2}\cos(Nx),$$

so that e_0, e_N are L^2-orthonormal, $\|e_0'\|_2 = 0$, and $\|e_N'\|_2 = N$. Put

$$\psi_1^{(N)} = \frac{e_N + e_0}{\sqrt{2}}, \qquad \psi_2^{(N)} = \frac{e_N - e_0}{\sqrt{2}}.$$

Choose any smooth $L^2((0, 2\pi))$-orthonormal functions ϕ_1, ϕ_2.

Theorem 1. *There is no finite-valued sequence $\gamma : \mathbb{N} \to [0, \infty)$ and no constant $C < \infty$ for which* (1)*, with $\lesssim$ replaced by $\le C$, holds for every $u \in H^1((0, 2\pi)^2)$ whose L^2-SVD converges in $H^1((0, 2\pi)^2)$ and every $r \ge 0$. The failure already occurs for smooth rank-two functions and $r = 1$.*

Proof. For $N \ge 1$, define

$$u_N(x, y) = 2\psi_1^{(N)}(x)\phi_1(y) + \psi_2^{(N)}(x)\phi_2(y). \tag{2}$$

This is smooth and has finite rank. Since the two $\psi_j^{(N)}$ and the two ϕ_j are L^2-orthonormal and the coefficients $2 > 1$ are distinct, (2) is its unique L^2-SVD up to signs. Hence

$$\sigma_1^{00} = 2, \qquad \sigma_2^{00} = 1,$$

and, at $r = 1$, the left side of (1) is

$$(\sigma_2^{00})^2 \|\psi_2^{(N)}\|_{H^1}^2 = 1 + \frac{N^2}{2}. \tag{3}$$

It remains to compute the $H^{(1,0)}$ singular values. The H^1 Gram matrix of the two left L^2 singular functions is

$$G_N = \begin{pmatrix} 1+N^2/2 & N^2/2 \\ N^2/2 & 1+N^2/2 \end{pmatrix}. \tag{4}$$

Indeed, the L^2 contribution is the identity, while both derivatives equal $e_N'/\sqrt{2}$.

As an operator $L^2((0,2\pi)) \to H^1((0,2\pi))$, u_N sends ϕ_1 to $2\psi_1^{(N)}$, sends ϕ_2 to $\psi_2^{(N)}$, and annihilates their orthogonal complement. Therefore the two nonzero squared $H^{(1,0)}$ singular values are the eigenvalues $\lambda_+(N) \geq \lambda_-(N) > 0$ of

$$A_N = \operatorname{diag}(2,1) G_N \operatorname{diag}(2,1) = \begin{pmatrix} 4+2N^2 & N^2 \\ N^2 & 1+N^2/2 \end{pmatrix}. \tag{5}$$

Now

$$\det A_N = 4(1+N^2), \qquad \lambda_+(N) \geq (A_N)_{11} = 4+2N^2.$$

It follows that

$$(\sigma_2^{10})^2 = \lambda_-(N) = \frac{\det A_N}{\lambda_+(N)} \leq \frac{4(1+N^2)}{4+2N^2} < 2. \tag{6}$$

There are no further nonzero singular values. Thus the right side of the purported bound at $r=1$ is at most

$$C\gamma(2)(\sigma_2^{10})^2 < 2C\gamma(2),$$

which is independent of N. This contradicts (3) as $N \to \infty$. $\square$

Remark 2 (Symmetric companion)**.** *Interchanging x and y gives the same counterexample to the analogous bound involving the $H^{(0,1)}$ singular values σ_k^{01} and the H^1 norms of the right L^2 singular functions.*

4 Verification, limitations, and novelty

The accompanying script verifies the Gram matrix, determinant, trace, the uniform bound $\lambda_- < 2$, and the diverging ratio for several exact values of N. These checks are not part of the proof.

The theorem settles the displayed question negatively under its natural uniform reading, even for smooth finite-rank functions on a fixed domain. It does not exclude estimates when γ or the comparison constant depends on the particular function, nor estimates under additional assumptions such as spectral localization or H^1-incoherence. Those nonuniform or restricted versions are consistent with the source's caveat "without further assumptions."

A bounded novelty search on 17 August 2026 covered the four run indexes, the local source, the current arXiv and journal records, the exact displayed question, title/arXiv-id queries, and Sobolev–SVD weighted-tail variants. No later answer or this rank-two Fourier family was found. Novelty confidence is moderate pending specialist review.

Human review recommendation. Check the normalization of e_N, the Gram matrix (4), the two-by-two strong-norm singular-value reduction, and the uniform interpretation of γ and the implicit constant.

References

[1] M. Ali and A. Nouy, *Singular Value Decomposition in Sobolev Spaces: Part II*, Z. Anal. Anwend. 40 (2021), 187–216; arXiv:1912.11293.